Data Sampling

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Section Introduction-Data Sampling

- What is Data sampling? Why Data Sample ?
- Supported types of Data sampling
- LAB Data sampling
- Summary.

What is Data Sampling?

Data Sampling is the process of selecting subset of data from very large tables with millions ,billions or trillions of rows with the goal of analyzing that data for insights without having to read all the data in the table.

Use Case : E-commerce seller Amazon sells over a 1.6 million products a day. That is about 30 million a month or 3.6 billion products a year. In 10 years it would be 36 billion rows of data. You were told to analyze data for 10 years to pick average number of sales a product in a day over the last 10 years.

How would you go about doing this ?

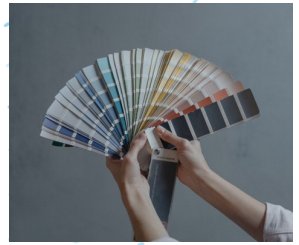
Option A: Would you count(*) WHERE product = 'Amazon Basics Rechargeable Battery' and divide the result by 365*10(PCT).

Option B: Would you just pick a random year in the last 10 years and do a count(*) WHERE product = 'Amazon Basics Rechargeable Battery' and divide the result by 365(PCT).

Option C: Would you count(*) randomly picked data from the table WHERE product = 'Amazon Basics Rechargeable Battery' divide the result by the number of rows returned(PCT)

Option A will be accurate but will consume lot of credits and will be slow.

Option B and Option C will be inaccurate.



What is Data Sampling?

Data Sampling is different from randomly pulling data from the table and analyzing it because when you randomly pull data from a table or pull data for a given year the data in the set is not representative of the data in the whole table

6 Red Apples = 0.54*100=54%
11 Population Apples

Random Data

1 Red Apples. = 0.25*100=25%
4 Sample Apples

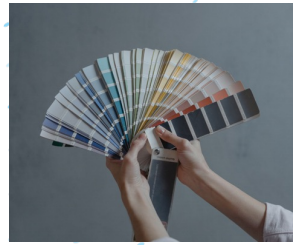
Sampling Data

2 Red Apples. = 0.50*100=50%
4 Sample Apples

What is Data Sampling?

Best Solution

Option D: Create a data sample from 10 years worth of sales data and conducting analysis on it would be the best approach to solve this data challenge.



Supported Types of Data Sampling.

Snowflake supports 2 types of Data sampling.

- ROW or BERNOULLI
- SYSTEM or BLOCK.

ROW/ BERNOULLI	SYSTEM/BLOCK.
Includes each row with probability of p/100. Eg p=5 means each row has 5% probability of being included. p is a probability value between 0 and 100.	Includes each block with a probability of p/100. Eg p=20 means each block/micro-partition has 20% probability of being included. p is a probability value between 0 and 100.
Row Sampling works on individual rows of a table reviewing each row to decide if it should be added to the sample. Row sampling is more accurate.	Block sampling is faster and consumes fewer credits as compared to row sampling. Block sampling works on blocks(micro-partition) and needs to consider only the inclusion or exclusion of a micro partition instead of looking at each row and is faster than ROW sampling.
Much slower than SYSTEM/BLOCK as every row needs to be examined, this is more true for very large tables with millions or rows. ROW/BERNOULLI works better with smaller tables	Block sampling is recommended for large tables with millions of rows. Block sampling on smaller tables can lead to inaccuracies due to smaller number of micro partitions and the data inside these partitions may not be representative.

Supported Types of Data Sampling.

- Number of rows generated by ROW/BERNOLLI sampling and SYSTEM/BLOCK sampling will be different as different algorithms are being used. Sample sizes will be close for medium to large tables.
- Use SEED to make the sampling deterministic. SEED can take a value from 1 to 2.147 billion(2,147,483,647)

Summary

- Data sampling is the process of selecting a subset of data from very large tables with millions, billions or trillions of rows with the goal of analyzing that data for insights without having to read all the data in the table.
- Extremely large tables benefit from Data sampling as smaller subset of data can be used for data analysis leading to time and credit saving.
- Snowflake supports ROW/BERNOLLI and SYSTEM/BLOCK of Data sampling.
- Sampled data accuracy will not be as high as running analytics on the entire population but will be close and sufficient for most use cases.
- Data sampling is different from randomly pulling data from a table, randomly pulled data is not a good representation of the data in the table and is not useful for analytics.
- ROW/BERNOLLI method works better with smaller tables.
- ROW/BERNOLLI method is slower for very large tables with millions of rows as all rows are reviewed before sample is generated.
- SYSTEM/BLOCK sampling is recommended for large tables as snowflake only needs to consider the inclusion or exclusion of entire micro-partitions of data, making it more efficient for sampling from extremely large tables.
- SYSTEM/BLOCK sampling is not recommended for smaller tables as smaller number of micro partitions results in a data skew with SYSTEM/BLOCK sampling.

Summary

- Data sample generated by the ROW/BERNOLLI method will be different from when generated by SYSTEM/BLOCK sampling as the algorithm used are different.
- Data sampling generates different set of rows every time they are run, both samples will still provide a good representation of the dataset from which they were derived, however rows selected will be different.
- We can use SEED to make the results deterministic. SEED can take any value from 1-2.147 billion(2,147,483,647)

Thank you

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